

common situation is that existing enterprise services are not designed for mobility.

Convertigo's MBaaS components address these requirements as follows:

- Platform transformation that enables the changing of existing internal services.
- Data filtering to expose only the required data model to mobile devices from the existing mobile services.
- Business logic augmentation to enhance existing internal services for mobile processes.
- These components are built on the very current concept called 'sequences', defined as 'steps' needed to get the data, transform it and handle it based on the business rules.

Creating sequences does not require any programming language; it is simply done by using object configuration in Eclipse based on the Convertigo studio GUI. The sequences can also be enhanced by using JavaScript steps to perform multi-faceted business logic.

Mobile security

An enterprise mobile platform must have security capabilities to secure data and processes.

These capabilities are:

- *User authentication* in order to be sure which user is accessing the enterprise data.
- *Access control* to control which part of the data should be seen for a particular mobile user.
- *Protocol encryption* to prevent network spies from reading data coming or going to the mobile devices. This is built on TLS 1.2 encryption and supports client and server certificates.
- *Mobile device data encryption* to prevent attackers from reading data on devices if they are lost or stolen.
- *Identity manager* to handle unique mobile identity and several different credentials to access the back-end system.

IoT integration

The Internet of Things (IoT), the next challenge facing enterprises, connects numerous devices at the same time. These devices take part in the enterprise ecosystem by interacting with existing systems and applications already deployed by companies.

Tracking objects, detecting smoke, monitoring machines and alerts for empty parking slots are some obvious use cases where IoT technology can help.

The IoT world connects a number of fields with each other in the following ways:

- Devices from IoT vendors are able to exchange data with network operators.
- IoT network operators deploy radio networks over the planet, which are able to handle IoT device traffic.
- Platform vendors provide software to connect the IoT network with active back-end systems or data repositories.

The Convertigo mobility platform is able to handle data coming for IoT devices through an IoT network operator. In this way, all the back-end services from Convertigo, such as connectors, sequences and security managers can be used to push and pull data from the back-end. The Convertigo mobility platform is fully integrated with the Sigfox network as a P3 platform.

The platform can be connected to the Sigfox network using the latter's call-back API. In this way, any IoT device triggering an event will be captured by the platform and will be processed by sequences to orchestrate activities to be done in the back-end systems.

Offline data capabilities

Nowadays, most of the mobile processing taking place, for example, at enterprises uses large files on the mobile devices. These files can be PDFs or some media files such as MPEG videos. It is easy to transfer large files online, but in offline mode, this is still difficult. This can be solved by using the Convertigo mobile platform, which provides a powerful file transfer device based on FullSync technology. So how exactly is this done? Basically, Convertigo can get a file from any enterprise control manager and transfer it to one or several mobile devices. The transfer is done in chunks so that if a file is not transferred totally when the network crashes, only the untransferred contents will be retransmitted. File transfers are done as background tasks as long as the mobile application is running. Convertigo provides these capabilities out-of-the-box as an SDK API that can be used on Android and iOS devices.

Mobile platforms are key components of the digital enterprise as most companies have designed their information systems for the Web and not for mobile devices. Mobile devices enable companies to gain agility, reduce development and maintenance costs, preserve system integrity and improve security.

With the increasing requirement for digitisation, enterprises will need constant mobile application development. The Convertigo mobility platform offers all the necessary components for enterprises to access the digital world, and for their customers, employees and their partners too — all with a control on costs and project timelines.

Using the Convertigo platform helps enterprises avoid having to redevelop the same services each time they need a new mobile application, hence slashing costs. 

Reference

<https://www.convertigo.com>

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